



4.5-1. Consider the following statements about linear programming and the simplex method. Label each statement as true or false, and then justify your answer.

- (a) In a particular iteration of the simplex method, if there is a tie for which variable should be the leaving basic variable, then the next BF solution must have at least one basic variable equal to zero.
- (b) If there is no leaving basic variable at some iteration, then the problem has no feasible solutions.
- (c) If at least one of the basic variables has a coefficient of zero in row 0 of the final tableau, then the problem has multiple optimal solutions.
- (d) If the problem has multiple optimal solutions, then the problem must have a bounded feasible region.

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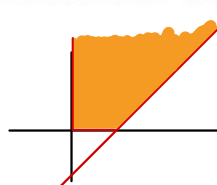
(a) TRUE. The ratio test tells how far the entering basic variable can be increased before one of the current basic variables drops below zero. If there is a tie for which variable should leave the basis, then both variables drop to zero at the same value of the entering basic variable. Since only one variable can become nonbasic in any iteration, the other will remain in the basis even though it will be zero.

(b) FALSE. If there is no leaving basic variable, then the solution is unbounded and the entering basic variable can be increased indefinitely.

(c) FALSE. All basic variables always have a coefficient of zero in row 0 of the final tableau.

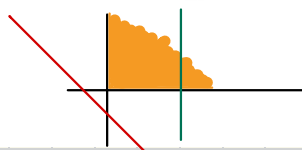
(d) FALSE.

Example 1: maximize $x_1 - x_2$
 subject to $x_1 - x_2 \leq 1$
 $x_1, x_2 \geq 0$



Clearly, any solution $(x_1^*, x_2^*) = (k + 1, k)$ for $k \in [0, \infty)$ with $z^* = 1$ is optimal. The problem has infinitely many optimal solutions and the feasible region is not bounded.

Example 2: maximize $-x_1$
 subject to $-x_1 - x_2 \leq 1$
 $x_1, x_2 \geq 0$



Any solution $(0, x_2^*)$ with $x_2 \geq 0$ is optimal.

4.5-5. A basic property of any linear programming problem with a bounded feasible region is that every feasible solution can be expressed as a convex combination of the CPF solutions (perhaps in more than one way). Similarly, for the augmented form of the problem, every feasible solution can be expressed as a convex combination of the BF solutions.

- (a) Show that *any* convex combination of *any* set of feasible solutions must be a feasible solution (so that any convex combination of CPF solutions must be feasible).
- (b) Use the result quoted in part (a) to show that any convex combination of BF solutions must be a feasible solution.

EX 3.5-3

3.5-3. Ed Butler is the production manager for the Bilco Corporation, which produces three types of spare parts for automobiles. The manufacture of each part requires processing on each of two machines, with the following processing times (in hours):

Machine	Part		
	A	B	C
1	0.02	0.03	0.05
2	0.05	0.02	0.04

Each machine is available 40 hours per month. Each part manufactured will yield a unit profit as follows:

	Part		
	A	B	C
Profit	\$300	\$250	\$200

Ed wants to determine the mix of spare parts to produce in order to maximize total profit.

FORMULATE THE LINEAR PROGRAM and FIND the SOLUTION USING THE SIMPLEX METHOD

Ex h.h.-P

$$\max \quad z = -x_1 + x_2 + 2x_3$$

$$\text{s.t.} \quad x_1 + 2x_2 - x_3 \leq 20$$

$$-2x_1 + 4x_2 + 2x_3 \leq 60$$

$$2x_1 + 3x_2 + x_3 \leq 50$$

$$x_1, x_2, x_3 \geq 0$$

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$$\max \quad z = 2x_1 + 3x_2$$

$$\text{st} \quad x_1 + 2x_2 \leq 20$$

$$x_1 + x_2 \leq 12$$

$$4x_1 + 3x_2 \leq 44$$

$$x_1, x_2 \geq 0$$

FIND the solution using the SIMPLEX METHOD

EX h.l.-S

$$\max \quad z = 5x_1 + 9x_2 + 7x_3$$

$$x_1 + 3x_2 + 2x_3 \leq 10$$

$$3x_1 + 4x_2 + 2x_3 \leq 12$$

$$2x_1 + x_2 + 2x_3 \leq 8$$

$$x_1, x_2, x_3 \geq 0$$